

Syllabus for

Academic year 2021/2022 >

MVE025 - Complex mathematical analysis

Komplex matematisk analys

Syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

6,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Chemical Engineering with Engineering Physics, Mathematics, Engineering Physics

Department: 11 - MATHEMATICAL SCIENCES

Teaching language: Swedish

Application code: 57136

Open for exchange students: No

Maximum participants: 130

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0105 Examination 6,0 c Grading: TH	6,0 c						Thu 28/10-2021 am J, Wed 05/01-2022 pm J, Fri 26/08-2022 pm J

In programs

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[David Witt Nyström](#)



[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Basic courses in mathematics.

Aim

To treat the fundamental theory for complex functions and to demonstrate important areas of application.

Learning outcomes (after completion of the course the student should be able to)

After completion of the course the student should be able to:

- define basic concepts and prove basic theorems in complex analysis,
- construct and analyse Möbius transformations and other conformal mappings,
- find Taylor and Laurent series of holomorphic functions,
- compute residues,
- calculate certain real integrals as well as contour integrals using residue calculus,
- use the Laplace transform and the z-transform to solve certain equations.

Content

Analytic and harmonic functions. Elementary functions and mappings. Multivalued functions, branch points. Complex integration. Cauchy's theorem. Cauchy's integral formula. Taylor and Laurent series. Isolated singularities. Residues.

Calculation of Fourier transforms using residues. Conformal mappings. Linear fractional mappings. Applications to the Laplace equation in the plane. The argument principle. Laplace and z-transforms and applications. Nyquist diagrams.

Organisation

Lectures and practical exercises.

Literature

Beck, Marchesi, Pixton and Sabalka: A First Course of Complex Analysis

See the course homepage for further information

Examination including compulsory elements

A written paper. Hand-ins can occur.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.